

READING DICTIONARY (v2) — the objects of D_IV⁵ and the KK→ruler descent, declared before derivation

A *dictionary*, not an atlas (Cal gate-v2 A). I delivered object→reading maps, not the transition maps between charts that make an atlas — so this is named a dictionary. The atlas (with transition maps between overlapping strata) is the *derived* goal that opens once this is gated; the transition maps are derivations, deferred. Declaration only — no derivations here. Keeper gates; derivation waits. Lyra = strata/operator side; @Grace = catalogue/data side (enumerated rows + kernel table). Reconnected + corpus-swept per entry (grep the proved theorem, not memory): T2566 (reading calculus), T2555/K1541/§505 (descent-circle split tier), K1359/K739/K1263 (U(1)_Y 3/13), T1949/T1938 (Pin(2)/SO(2)), T2562/K1487 (Kostant Dirac), T2569 (ruler), Elie 5410 ($\mathbb{Z}_2$ distinctions).

0. The reading calculus R — declared FIRST (Cal gate-v2 B)

A reading is the value of a declared verb applied to a declared object (Section 2). The verbs R: (i) isotypic decomposition under a subgroup; (ii) Casimir eigenvalue $C_2(\lambda)$; (iii) Frobenius–Schur reality-type ($\mathbb{R}/\mathbb{C}/\mathbb{H}$); (iv) kernel evaluation / residue / norm; (v) boundary-orbit rank / support-stratum index; (vi) Peirce grading; (vii) generator count. A quantity is a “reading” iff its (object, verb) pair is one Section-2 entry $\times$ one verb above — nothing else counts. Every result is a reading of *one operator* on D_IV⁵; never “commitment forces D_IV⁵.”

1. Completeness criterion — up front (Cal gate-v2 C)

The object-classes are exhaustive because they are the *complete invariant data* of a bounded symmetric domain, organized by four complete decompositions that between them cover all six entries — a claim stated to be gated, not buried: - the G-orbit stratification of the closed domain $\bar{D}$ (Korányi–Wolf) is finite and complete: the open bulk orbit (the interior — entry 1) *plus* the rank + 1 = 3 boundary strata (entry 3). The bulk is not a separate kind — it is the *open* (generic, top-dimensional) G-orbit, and the boundary strata are the lower orbits; bulk and boundary together are *all* the G-orbits, no more. - the KAK (polar / Cartan) decomposition $D = K \cdot A \cdot o$ is complete (every point is KAK): its radial part is the Cartan slice A, the maximal flat (entry 2), and its angular part is the K-type basis (entry 5). The Cartan slice is not a separate kind — it is the “A” half of the KAK harmonic analysis whose “K” half is the K-types; together they coordinatize the domain. - the Jordan/Peirce decomposition is complete for a rank-2 Jordan triple: $V = \mathbb{R}c_1 \oplus V_{12} \oplus \mathbb{R}c_2$ (entry 4), exhaustive; - the reproducing kernels are the two canonical ones — Bergman (bulk) and Szegő (Šilov) (entry 6); a bounded symmetric domain has exactly these two. Completeness claim: any geometric object a physical reading can attach to is a G-orbit (bulk or boundary stratum), a KAK component (Cartan slice or K-type), a

Peirce block, or a reproducing kernel — these four complete decompositions cover all six entries of Section 2, and no object-class lies outside them. (Gate this claim.)

2. The dictionary entries — object $\rightarrow$ defining datum $\rightarrow$ source $\rightarrow$ sweep (Cal gate-v2: sweep per entry)

#	object-class	defining datum	corpus source	sweep note
1	bulk (open interior)	Bergman metric + kernel $K_B(z,w)$; Kähler form	Faraut–Korányi; T2569	swept: bulk = the D_{IV}^5 interior, Bergman-metric complete
2	Cartan slice (maximal flat)	two Cartan coords (KAK polar); B_2 roots; $W(B_2)$	$\rho = (5/2, 3/2)$, T2517	swept: rank-2 flat; ρ -vector banked
3	Šilov boundary strata (Korányi–Wolf flag)	boundary-orbit flag, rank+1 = 3 strata; top $\check{S} = (S^4 \times S^1)/\mathbb{Z}_2^{\wedge} \check{S}$	F666, T2517 (addresses $\{5/2, 3/2, 0\}$)	swept: 3 strata = the support flag; do not confuse with the retracted genus- $\{5, 3, 0\}$
4	Peirce blocks (Jordan)	$V = \mathbb{R}c_1 \oplus V_{12} \oplus \mathbb{R}c_2$; dims (1, 3, 1); Peirce grading $\mathbb{Z}_2^{\wedge} \text{Peirce}$	Jordan Peirce; census $(\mathbb{C}, \mathbb{H}, \mathbb{R})$	swept: $(1, n-2, 1) = (1, 3, 1)$; V_{12} real $SO(3)$ -vector (T2545)
5	K-types $(SO(5) \times SO(2))$	highest-weight lattice (a, b) ; $C_2(\lambda) = \langle \lambda + \rho, \lambda + \rho \rangle - \langle \rho, \rho \rangle$; lowest $C_2 = 6$	Wallach 1976; $C_2 = 6$	swept: Casimir line; spinor block = the $\mathbb{H}$ K-type
6	Bergman–Szegő kernels	K_B (bulk), K_S (Šilov), Hardy H^2 ; Gindikin Γ_{Ω} ; $c_{FK} \cdot \pi^{(9/2)} = 225$	Hua; F157 (Γ_{Ω}) ; $\ f_0\ ^2 = \Gamma(5/2)^2 / \Gamma(1/2)$	swept: two canonical kernels; Γ_{Ω} is the object's own kernel re-read (not a new operation)

3. The $\mathbb{Z}_2$ carriers — subscript EVERY one (Cal gate-v2 F; Elie 5410 retraction ratified)

There is one physical non-orientability (I_2 of the theorem), presented by several distinct objects. Do not merge; subscript: - $\mathbb{Z}_2^{\check{S}}$ — $w_1 \in H^1(\check{S}; \mathbb{Z}_2)$, the Šilov double-cover deck group (the theorem's twist generator; *consequence*: boundary carries Pin^- , not Spin — name the cause, not the consequence); - $\mathbb{Z}_2^{\text{Möb}}$ — the Möbius-locus w_1 (K3 / $\text{Pin}(2)\text{-}\mathbb{Z}_2$, T1949; $\text{SU}(2)_L$ couples only to LH windings here); - $\mathbb{Z}_2^{\text{orient}}$ — the orientation quotient $\text{Pin}(2)/\text{SO}(2) = \mathbb{Z}/\text{rank} = \mathbb{Z}_2$ (T1938/T1949 — Elie's *surviving, ratified* reading); - $\mathbb{Z}_2^{\text{Peirce}}$ — the Jordan-frame grading $c_1 \leftrightarrow c_2$ (a different $\mathbb{Z}_2$; *not* a non-orientability carrier — do not file it with the three above).

Open (Grace flag): the carrier relationship among $\mathbb{Z}_2^{\check{S}}$, $\mathbb{Z}_2^{\text{Möb}}$, $\mathbb{Z}_2^{\text{orient}}$ is not proved — three presentations that may or may not coincide; banked as distinct until a map is derived. (*Elie's $\text{Pin}(2)$ retraction, ratified: his checks attacked a name-reconstructed $\text{Pin}(2)$; T1949's $\text{Pin}(2)/\text{SO}(2)=\mathbb{Z}_2$ orientation quotient is the reading that survives. This is the repo-grep rule catching the person applying it — the system working.*)

4. The $\text{KK} \rightarrow \text{ruler}$ descent — objects declared, TIERS carried (Cal gate-v2 D)

Ruler = one input, already closed (T2569, inherited not re-derived): scale is a forced necessity (dimensional-bar / Casimir-runaway — GR-level single input); shape (S^4/S^1 ratio) is not forced (Šilov boundary at infinite Bergman distance $\rightarrow$ conformal/CR, not a metric).

The descent objects (declared; tiers cited to the LINE, not the title): - the circle — the $\text{SO}(2)$ factor of $K = \text{SO}(5) \times \text{SO}(2)$. ★ TIER (T2555 split, Cal §505 / K1541 — cite this line, never the T2555 title): the totally-real half-dim reduction bulk-10 $\rightarrow$ boundary-5 is DERIVED; the $S^1 = \text{SO}(2)$ drop, boundary-5 $\rightarrow$ 4, is STRUCTURAL, NOT DERIVED — a *separate* reduction resting on an owed one-bit-per-dimension primitive (T2556). $\Rightarrow$ the $\text{SO}(2)$ circle must not serve as a derived constraint for the descent; carry it as structural. - the chain — $\text{SO}(5,2) \supset \text{SO}(4,2) \supset \text{SO}(3,1)$ (conformal $\rightarrow$ Lorentz; Casey #14 signature). *Declared object.* - the 4D target — 4D Minkowski as the $\text{SO}(3,1)$ Lorentz stratum. *Declared.* - the operator-region — the Kostant cubic Dirac D on the spinor sector (self-adjoint to 10^{-15} in the FK metric, discrete spectrum $\rightarrow \infty$; T2562/K1487). NOT Hilbert–Pólya (retracted, Elie 5286 — power-law Weyl $\neq$ logarithmic Riemann); do not re-open.

5. Open rows — declared, not derived

open row	the question	corpus	lean / status
U(1)_Y origin — RULED (K1359 ratified)	trace-vs-KK fork — ruled: trace- normalization (of End_K / the census), no KK win	K1359 (ruled + ratified, Cal cold-read); c_5/c_3 = 3/13 exact Chern math but the $\sin^2\theta_W$ identification is unforced + $M\bar{S}(M_Z)$ - scheme- dependent, RETIRED as a running coincidence (K739/K1263)	RULED, not open: 3/13 stays IDENTIFIED. One honest open piece for a <i>future</i> <i>note</i> (Cal): why trace- normalization is the right one in a framework whose headline is <i>no</i> <i>unification</i> — “it’s what NCG does” is not enough
no- ν_R carrier — RESOLVED	$\mathbb{Z}_2^{\wedge}\text{orient}$ (the orientation quotient $\text{Pin}(2)/\text{SO}(2)$, an <i>involution</i>) carries the “ ν_R cycle can’t close” reading	T1949 (Möbius/involution mechanism), T1984 (sterile- ν exclusion); Elie 5411 ($\mathbb{Z}_2$ inventory — orientation $\mathbb{Z}_2$ ’s don’t all live in H^1)	resolved: carrier = $\mathbb{Z}_2^{\wedge}\text{orient}$ (involution). Drop the Šilov separator — ν_R non-closure rides on the orientation <i>involution</i> , not Šilov-orientability; a $\mathbb{Z}_2^{\wedge}\bar{S} / D_{IV}^{7-8}$ separator is <i>invalid</i> for this row
$\mathbb{Z}_2$ -carrier coincidence	do $\mathbb{Z}_2^{\wedge}\bar{S}$, $\mathbb{Z}_2^{\wedge}\text{Möb}$, $\mathbb{Z}_2^{\wedge}\text{orient}$ coincide?	§3	open — distinct until a map is proved
S^1 -drop primitive	the one-bit-per- dimension bridge for boundary-5→4	T2556 (owed)	owed — blocks the $\text{SO}(2)$ -circle- derived route

Gate ask (Keeper, v2)

Confirm before derivation: (A) named a dictionary — transition maps (the atlas) deferred as derivations; (B) reading calculus R declared first; (C) completeness criterion up front; (D) descent tiers carried — $\text{SO}(2)$ -circle drop is STRUCTURAL (cite §505/K1541, not the T2555 title), cannot be a derived constraint; (E) the **U(1)_Y trace-vs-KK fork (K1359)** carried as the live open row with the “no win, 3/13 Identified” lean; (F) every $\mathbb{Z}_2$ subscripted (four carriers, $\mathbb{Z}_2^{\wedge}\text{Peirce}$ filed separately); plus

a corpus sweep per entry and the no- ν _R carrier as an open row. On Keeper-PASS, derivation opens (Elie’s Bergman/Szegő strata backbone ready).

★ KEEPER GATE VERDICT (K1757, 2026-08-21) — CONDITIONAL PASS

All six of Cal’s gate-v2 sharpenings are folded and GREEN: A0 named a dictionary (atlas/transition-maps deferred); B reading calculus R declared FIRST (seven verbs, “iff (object,verb), nothing else”); B5/F four $\mathbb{Z}_2$ ’s subscripted with the carrier-coincidence OPEN; C completeness criterion up front; C2/D descent tiers carried to the LINE (SO(2)-drop STRUCTURAL, §505/K1541 not the T2555 title); D1 Kostant Dirac scoped to T2562, Hilbert–Pólya kept retracted; E the K1359 U(1)_Y trace-vs-KK fork carried as the live open row with the “no win / 3/13 Identified” lean; corpus-sweep per entry; no- ν _R open row present. Strong work — this is the object *declared*, not derived.

ONE gate finding (why CONDITIONAL, not full PASS) — the completeness criterion (Section 1, the load-bearing item): Section 1 argues exhaustiveness over four object-kinds (boundary strata / Peirce blocks / K-types / reproducing kernels), but Section 2 declares SIX objects — the bulk (object 1) and the Cartan slice (object 2) are not covered by the Section-1 completeness argument. Fix (pick one): (a) extend the completeness claim to include them (bulk = the domain interior itself, the KAK “A”-and-“N”/radial datum; Cartan slice = the maximal flat, the “A” of KAK — both canonical BSD data), so the exhaustiveness statement ranges over all six; or (b) show bulk + Cartan slice as derived-from / special-cases-of the four kinds. Either closes it; a completeness claim must match the actual list, or a referee asks exactly this.

One MINOR (not gating, enforce at derivation time): the six objects are *definitional* (they exist), so B3’s forced-vs-measured D/PD/I/C/S tier attaches to each *reading* when it is derived, not to the object-classes here — carry the tier column into the derivation phase, one tier per reading.

Two corpus facts to fold before derivation opens (K1757): 1. T1949/T2328 reconciled (Keeper adjudication): the no- ν _R Möbius non-orientability is carried by the involution $\mathbb{Z}_2^{\wedge}\text{orient}$ (H^0), NOT a w_1 — the locus’s cover is a contractible orientable 5-ball (T2328), non-orientable only via the deck involution. So the no- ν _R carrier open row RESOLVES to $\mathbb{Z}_2^{\wedge}\text{orient}$, and the Šilov w_1 separator ($\mathbb{Z}_2^{\wedge}\check{S}$, H^1) cannot test it (different mathematical kinds). Update Section 5’s no- ν _R row: carrier = $\mathbb{Z}^{\wedge}\text{orient}$ (involution); the shipped table’s Šilov separator for that row is invalid → drop it / re-key to the involution. 2. K1359 RULED (Cal §652, ratified K1757): route (B)/KK is DEAD — 3/8 is a high-scale ($\sim 10^{13}$ GeV) boundary value, the measured 0.23077 is the M_Z value; different TYPE, not competing values; route (A) is not normalization-free. The lean holds: $3/13 = N_c/(N_c+2n_C)$ stays IDENTIFIED. Keep the row, mark it ruled-not-open (the fork settled without the proposed com-

putation).

On the completeness fix (one edit) $\rightarrow$ PASS $\rightarrow$ derivation opens. Keeper + Cal + Casey GO to open the derivation phase per standing.

****Lyra + Grace, 2026-08-21 (gate-v2 re-draft). Renamed to a DICTIONARY (object $\rightarrow$ reading; transition maps = the atlas, deferred). R declared first; completeness criterion up front; six object-classes corpus-swept; FOUR $\mathbb{Z}_2$'s subscripted ($\mathbb{Z}_2^{\text{Š deck}} / \mathbb{Z}_2^{\text{Möb}} / \mathbb{Z}_2^{\text{orient}} = \text{Pin}(2)/\text{SO}(2)$ T1949 / $\mathbb{Z}_2^{\text{Peirce}}$ filed apart) with the carrier-coincidence OPEN; descent tiers carried explicitly (SO(2)-drop STRUCTURAL not derived, cite §505/K1541 not the T2555 title); the U(1)_Y trace-vs-KK fork (K1359) is the live open row, lean = no win / 3/13 stays Identified; no- ν_R carrier open. Keeper gates; derivation waits. Every result a reading of one operator on D_IV⁵. Nothing pushed; CP existence-only.****